Topic - 7 :

Encapsulation:

Encapsulation is the process of wrapping data (variable) and methods (function) together inside a class and restricting direct access to some of the data to protect it..

Encapsulation = Data hiding + Controlled access.

Instead of allowing direct access to variable, we will

- hide data
- provide controlled methods to access/modify it

Ex
a bank account

You cannot directly access or change : $\boxed{\text{account balance}}$

- Instead you use:
  - deposit()
  - withdraw()

So, Data (balance) → hidden

Access → controlled through methods

This is encapsulation.

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Why Encapsulation needed.

without encapsulation:

- anyone can change data.
- lead to bugs
- security issues....

with encapsulation:-

* Data is protected
* only valid operation allowed..

Example without Encapsulation:-

```
class BankAccount:
    def __init__(self, balance):
        self.balance = balance

acc = BankAccount(1000)
acc.balance = -500 # problem [changing w/o security chk].
print(acc.balance) # 500
```

without performing Deposit or withdrawing it has been changed so that

No security.

* Everyone can access them so that it will be easy...

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< Encapsulation using private variable >:

In python, we make variable private by using

__ (double underscore).

Eg

```
Class BankAccount:
    def __init__(self, balance)
        self.__balance = balance # private variable

    def deposit(self, amount):
        if amount > 0:
            self.__balance += amount.

    def withdraw(self, amount):
        if 0 < amount <= self.__balance:
            self.__balance -= amount

    def get_balance(self):
        return self.__balance
```

use

```
acc = BankAccount(1000)
acc.deposit(500)
acc.withdraw(200)
print(acc.getBalance())  # 1300
```

If `print(acc._balance)` ✗

Types of access in python

① public
self.name
→ accessible anywhere

② protected
self._name
→ should be used carefully.

③ private
self.__name
→ strictly restricted...

Real world

Encapsulation used in
- Banking system
- APIs
- Authentication systems
- Payment gateways..

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